YAYUN (BIRDY) CHEN

Department of Agricultural Economics
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RESEARCH INTERESTS

Environmental and Energy Economics; Natural Gas, Hydrogen, Electricity, and Bioenergy Markets; Energy Transition, Decarbonization, and Environmental Policy Design; Climate Risk and Resilience; Applied Econometrics and Optimization-Based Market Modeling

EMPLOYMENT

Associate Research Scientist, Texas A&M University 2024- present
Postdoctoral Research Associate, Texas A&M University 2024
Gas Market Analyst, RBAC Inc. 2019–2020
Developed the G2M2 global natural gas market model and its user interface; natural gas market analysis and demand modeling.

EDUCATION

Ph.D. in Agricultural Economics, Texas A&M University 2020-2024
Dissertation: Impacts of Hurricanes and Algae-based Jet Fuel on Energy Production Economics and Markets
Advisors: James Mjelde & Bruce A. McCarl
M.A., Economics, The University of Texas at Austin 2019
M.A., Economics, National Tsing Hua University 2016
B.A., Economics (with Honors), Lanzhou University 2013

PEER-REVIEWED PUBLICATIONS

Energy and Climate

[10] **Chen, Y.**, X. Deng, C. Fei, B.A. McCarl, and Y. Xu. 2025. "Forecast-Based Disruptions in the Energy Sector: Economic Impacts of Anticipatory Hurricane Responses in U.S. Gulf Coast Refineries." *Environmental Research Letters*, 20, 114025. https://doi.org/10.1088/1748-9326/ae0e3a

[9] Lin, N., **Y. Chen**, and M.P. Madariaga. 2024. "Route-to-market strategy for low-carbon hydrogen from natural gas in the Permian Basin." *Fuel*, 355: 129420. https://doi.org/10.1016/j.fuel.2023.129420

Bioenergy and Low-Carbon Technologies

[8] Dhatt, P., W. Li, **Y. Chen**, X. Meng, M. Liu, C. Fei, S.Y. Dai, and J.S. Yuan. "Bioinspired and Biodegradable Fluorine-Free Superhydrophobic Coatings." *Matter*, Accepted.

[7] Wei, W., Q. Hu, **Y. Chen**, Y. Kwon, S. Xi, M. Liu, K. Chen, W. Li, C. Fei, S.Y. Dai, Q. Chang, and J.S. Yuan. "Sustainable and Efficient Protein Production by Electro-biomanufacturing Enabled by Co-design of Catalysts and Bioprocesses." *One Earth*, Accepted.

[6] Li, W., C. Hu, T. Zhu, **Y. Chen**, R. Zhang, P. Jiang, P.S. Dhatt, J. Li, S. Xiang, M. Liu, C.J. Fei, J. Liu, S.Y. Dai, and J.S. Yuan. 2026. "Transforming Renewable Carbon Fiber Performance, Economics, and Sustainability via Oriented Crystallization Design." *Matter*, 103001. https://doi.org/10.1016/j.matt.2026.103001

[5] Racine, C.J., A. Badreldin, J. Pellessier, **Y. Chen**, S. Chen, S. Wang, J. Feng, C. Fei, Y.H. Hu, and Y. Li. "Large-Scale Synthesis (75 g/batch) of Single-Atom Catalysts for Selective Electrochemical CO_2 Reduction to CO and Commercialization Potential Analysis." *ACS Omega*, https://pubs.acs.org/doi/10.1021/acsomega.6c02693.

[4] Bin, L., Q. Li, C. Hu, **Y. Chen**, Y. Zeng, S. Pearson, M. Liu, C. Fei, J.S. Yuan, and S.Y. Dai. 2025. "Remediation and upcycling of microplastics by algae with wastewater nutrient removal and bioproduction potential." *Nature Communications*, 16, 11570. https://doi.org/10.1038/s41467-025-67543-5

[3] **Chen, Y.**, K. Chen, C. Fei, S.Y. Dai, and J.S. Yuan. 2025. "Life Cycle Assessment and Social Benefits of Producing Bioplastics via Integrating Electrochemical CO_2 Conversion and Microbial Fermentation." *ACS Sustainable Chemistry & Engineering*, 13, 21367-21379. https://pubs.acs.org/doi/10.1021/acssuschemeng.5c02891

[2] Zhao, R., W. Li, J. Gonzalez-Aguirre, **Y. Chen**, J.S. Yuan, H. Pakrasi, C. Fei, S. Park, G. Roell, Y. Tang. 2025. "Development of Multimodal AI for Photobiorefineries via Knowledge Syntheses, Transfer Learning, and Techno-Economic Analysis." *Bioresource Technology*, 133646, ISSN 0960-8524. https://doi.org/10.1016/j.biortech.2025.133646.

[1] Chen, K., P. Zhang, **Y. Chen**, C. Fei, J. Yu, J. Zhou, Y. Liang, W. Li, S. Xiang, S.Y. Dai, and J.S. Yuan. 2025. "Electro-biodiesel empowered by co-design of microorganism and electrocatalysis." *Joule*, 9(1), 101769. https://doi.org/10.1016/j.joule.2024.10.001

Working papers available upon request.

FUNDING

As PI or Co-PI

[1] "A Kinetic–Environmental–Economic Simulation Platform (KEEP) to Guide Biomanufacturing Experiments and Supply-Chain Decisions for CO2–Derived Products." CURB Engineering Research Center (NSF) (Trainee Grants, PI, 2026–2027, $15,000).

[2] "Climate Change and Global Agriculture." USDA–Foreign Agricultural Service (Co-PI, 2024–2027, $2,500,000, 20% share).

[3] "The Impact of English Proficient Reclassification on Long-Term Educational and Earnings Outcomes." University of Houston ERC Project #055 (Co-PI, 2023–2027).

TEACHING EXPERIENCE

University of Missouri

Invited Lecturer, Green Engineering (BIOL-EN/CH-ENG 4001/7001) Spring 2026

Energy Institute, Texas A&M University & King Abdullah Petroleum Studies and Research Center (KAPSARC)

Co-Designer and Instructor, Energy, Policy and Environment Nov. 2023

Texas A&M University

Lab Instructor, AGEC 641 Applied Mathematical Programming (PhD level) Spring 2023

Teaching Assistant, AGEC 350 Natural Resource and Environmental Economics 2020–2021

Teaching Assistant, AGEC 621 Econometrics for Agribusiness (Master level) 2021

Teaching Assistant, AGEC 344 Agricultural and Food Law 2020–2021

National Tsing Hua University

Teaching Assistant, Global Economy (EMBA) 2015

AWARDS AND HONORS

AAEA Travel Grant for Early Professionals 2026

US Association for Energy Economics, Young Professional Best Paper Award, Finalist 2024

Department Scholarship, Department of Agricultural Economics, TAMU 2024

Travel Scholarship (AAEA), Department of Agricultural Economics, TAMU 2022

GSA Travel Grant, Department of Agricultural Economics, TAMU 2021

Graduate Merit-based Tuition Scholarship, Department of Economics, UT Austin 2018-2019

Graduate Student Full Scholarship, NTHU, Taiwan 2013-2016

Lanzhou University Outstanding Undergraduate Honor 2013

Lanzhou University Outstanding Student Scholarship 2010-2012

Lanzhou University Outstanding Leadership Honor 2010-2011